Assignment_Linear Transformation_Dual Spaces

Let V be an n-dimensional vector space over R (the discussion below goes for any field K). A linear transformation from V into R is also called a linear functional. Denote the vector space of all linear transformations from V into R by V^*. $V^*$is an n-dimensional vector space. It is called the dual space of V. Given an ordered basis of V, any vector v is represented by a nx1 column vector. The matrix representation of any linear functional in $V^*$is an 1xn matrix, a row vector.

Let $\{v_1,\ldots,v_n\}$ be a basis of V. A basis $\{f_1,\ldots,f_n\}$ of $V^*$is called the dual basis of $\{v_1,\ldots,v_n\}$ whenever, $f_i(v_j) = \delta_{ij}$, where $\delta_{ij}=1$ when i=j and 0 otherwise.

(1) Given a basis $\{v_1,\ldots,v_n\}$ of V, show the existence and uniqueness of a dual basis $\{f_1,\ldots,f_n\}$ of V^*.

The vector space of all linear functionals on V^*, denoted by V^{**}, is again an n-dimensional vector space. For any v in V, we define e(v) as a linear functional on V^*such that for any f in V^*, e(v)(f) = f(v).

(2) Show that e: V→$V^{**}$is an isomorphism.

(3) Let $\{f_1,\ldots,f_n\}$ be a basis of V^*. Show that there exists a basis $\{v_1,\ldots,v_n\}$ of V such that $\{f_1,\ldots,f_n\}$ is its dual basis.

Let T: V→W be a linear transformation. Then, we can define

T^*: $W^* \to V^*$ such that $T^*(g)(v)=g(T(v))$. T^* is called the transpose of T.

(4) Let β, γ be ordered basis of V, W respectively. Then, let β', γ' be the dual bases of β, γ respectively. Show that the transpose of $[T^*]_{\beta', \gamma'}$ is equal to $[T]_{\gamma}^{\beta}$, i.e. $([T^*]_{\beta', \gamma'})^T = [T]_{\gamma}^{\beta}$

(5)Show that Rank T= dim ImT=Rank T^*= dim ImT^*

(i.e. column rank = row rank)

(Hint: Suppose dim ImT= k and let $\{w_1,\ldots, w_k\}$ be a basis of ImT. Extend $\{w_1,\ldots, w_k\}$ to $\{w_1,\ldots, w_k, \ldots, w_m\}$, a basis of W.

Let $T(v_1) = w_1,\ldots, T(v_k) = w_k$. Then, $\{v_1,\ldots, v_k\}$ is linearly independent. Let $\{v_{k+1},\ldots, v_n\}$ be a basis of Ker T. Then, $\{v_1,\ldots, v_k\} \cup \{v_{k+1},\ldots, v_n\}$ is a basis of V. Let $\{w^*_1,\ldots, w^*_k, \ldots, w^*_m\}$ be the dual basis of

$\{w_1,\ldots, w_k, \ldots, w_m\}$.

Show that (i)$\{T^*(w^*_1),\ldots,T^*(w^*_k)\}$ is linearly independent.

(ii) $T^*(w^*_{k+1})=\ldots=T^*(w^*_m) = 0$. Hence, $\{T^*(w^*_1),\ldots,T^*(w^*_k)\}$ is a basis for ImT^* and then dim ImT^*=k)